

Unforced Finite-Time Blow-Up for Navier–Stokes: Exact Nonlinear Closure on the Three-Dimensional Periodic Torus

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Abstract

We study whether the smooth forcing in a periodic finite-time Navier–Stokes blow-up carrier can be removed without destroying the singular asymptotics. A support-preserving core parametrix is upgraded by Neumann inversion, coupled to a global periodic regular inverse, and closed under the quadratic nonlinearity in a graph norm with one derivative in reserve. Under the source-backed estimates stated in the paper, the resulting late-slab correction solves the unforced equation while remaining asymptotically smaller than the carrier, thereby preserving its leading blow-up coefficient.

Proof- and formalization-status note. The forced carrier and numerous local analytic identities are imported from the released OpenAI manuscript and its Lean repository [1, 2]. This release includes a concrete, placeholder-free `UnforcedClosure.lean` source file for the abstract closure layer, together with a pinned Lean/Mathlib manifest, exact-rational replay, source-hygiene checks, SHA-256 hashes, and reproducibility receipts. The abstract closure source has now been kernel-compiled successfully in the pinned Lean/Mathlib environment: `lake env lean UnforcedClosure.lean` and the local `lake build` pass, while the exact-arithmetic/source-hygiene replay and Pytest suite also pass. The PDE-specific hypotheses in Appendix A have not yet been discharged by a compiled end-to-end theorem against the frozen upstream repository. Accordingly, the machine-checked result presently covers the abstract closure layer, not the full unforced Navier–Stokes theorem. Priority language remains qualified pending PDE-specific Lean integration, independent source replay, and refereeing.

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1 Problem, setting, and main theorem

1.1 The unforced periodic problem

We work on the unit three-torus

$$\mathbb{T}^3 = \mathbb{R}^3 / \mathbb{Z}^3,$$

with viscosity $\nu > 0$. The unforced incompressible Navier–Stokes equations are

$$\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = 0, \tag{1.1}$$

$$\operatorname{div} u = 0. \tag{1.2}$$

The classical periodic problem asks whether every smooth divergence-free datum generates a smooth solution for all positive time. The official Clay formulation distinguishes global smoothness with $f \equiv 0$ from breakdown alternatives allowing a smooth force [3]. On 8 September 2026, OpenAI released a manuscript and accompanying Lean formalization claiming finite-time breakdown with smooth forcing on both $\mathbb{R}^3$ and $\mathbb{T}^3$ [1, 2]. On 11 September 2026, the Clay Mathematics Institute stated that the problem had “apparently been settled,” while emphasizing that its formal evaluation and credit process would proceed under the published prize rules and was deliberately unhurried [4]. Accordingly, throughout this paper the released OpenAI construction and frozen repository are treated as pinned mathematical source inputs, not as an independently accepted resolution of the Millennium Prize Problem. The question addressed here is whether the smooth forcing in that released construction can be removed while retaining the same singular carrier.

The historical framework begins with Navier and Stokes [5, 6] and the inviscid precursor of Euler [7]. Leray’s global weak solutions [8] and Hopf’s weak-solution theory [9] established the basic finite-energy setting. Regularity criteria include Prodi and Serrin [10, 11], the scale-critical result of Escauriaza–Seregin–Šverák [12], and partial regularity of Caffarelli–Kohn–Nirenberg [13]. Classical local and small-data theory

is represented by Fujita–Kato and Kato [14, 15], with modern functional-analytic treatments in [16–21].

The singular-flow context includes Tao’s averaged model [22], convex-integration nonuniqueness for weak Navier–Stokes solutions [23], and forced Leray nonuniqueness [24]. Oscillatory and instability mechanisms used in the underlying forced carrier have antecedents in Lifschitz–Hameiri [25], Friedlander–Vishik [26], Craik–Criminale [27], Singh–Sridhar [28], Leibovich–Stewartson [29], and Billant–Gallaire [30, 31]; stress realization has links to Euler convex integration [32]. Recent singularity constructions with forcing or modified dissipation include [33–35]. The author’s earlier unforced precursor is deposited as Zenodo record 22857336 [36].

1.2 Imported periodic forced carrier

The OpenAI construction gives, after rescaling and periodization, a smooth periodic solution (U_*, P_*) on $[0, 1)$ with a smooth force F_* ,

$$\partial_t U_* + (U_* \cdot \nabla) U_* - \nu \Delta U_* + \nabla P_* = F_*, \quad (1.3)$$

$$\operatorname{div} U_* = 0. \quad (1.4)$$

The carrier is smooth for each $t < 1$, periodic, and its velocity and pressure can be placed in a fixed compact subset of the interior of a fundamental cube. The periodization is exact because distinct integer translates have disjoint support; in particular the nonlinear transport term periodizes without cross-terms [1, Cor. 10.6]. The repository comparator theorem likewise records the forced periodic breakdown statement [2].

Near the distinguished blow-up path x_τ , $\tau = 1 - t \downarrow 0$, the leading azimuthal velocity has the form

$$U_{*,\theta}(1 - \tau, x_\tau) = \tau^{-A}(e_0 + O(\tau^{2h})), \quad A = \frac{1}{2} + h, \quad e_0 > 0, \quad (1.5)$$

with $0 < h < 1/2$; the OpenAI construction uses a much smaller fixed range for h , which is contained in this interval [1].

1.3 Main analytic release theorem

Let

$$\mathcal{N}_\nu(u, p) := \partial_t u + (u \cdot \nabla) u - \nu \Delta u + \nabla p. \quad (1.6)$$

The new problem is to find (v, π) on a late slab $[t_0, 1)$ such that

$$\mathcal{N}_\nu(U_* + v, P_* + \pi) = 0. \quad (1.7)$$

Expanding about the carrier gives

$$L_* v + \nabla \pi + F_* + B(v, v) = 0, \quad (1.8)$$

where

$$L_* v := \partial_t v - \nu \Delta v + (U_* \cdot \nabla) v + (v \cdot \nabla) U_*, \quad (1.9)$$

$$B(v, w) := (v \cdot \nabla) w. \quad (1.10)$$

The inverse construction below is carried out in a pressure-treated source space: equivalently, one may regard the linear operator as acting on the solenoidal velocity correction together with its zero-mean

pressure component, or eliminate that pressure component and work with the corresponding solenoidal reduction. All identities of the form $LG = I$ are identities in this pressure-treated source space. The support statements in the core branch refer to the velocity component and are not inferred from a nonlocal global Leray projection. Section 7 reconstructs the physical pressure explicitly and thereby returns to the unprojected equation (1.8).

Theorem 1.1 (Unforced late-slab closure). *Fix $\nu > 0$ and a periodic forced blow-up carrier (U_*, P_*, F_*) satisfying (1.3)–(1.5). Assume the source-backed core parametrix estimates, valid-band gluing identities, support bounds, and regular periodic inverse properties stated precisely in Appendix A. Then there exists $t_0 < 1$, a smooth periodic divergence-free correction v , and a smooth periodic pressure correction π on $\mathbb{T}^3 \times [t_0, 1)$ such that*

$$\mathcal{N}_\nu(U_* + v, P_* + \pi) = 0.$$

Moreover the correction is asymptotically subleading along the distinguished path,

$$\tau^A v_\theta(1 - \tau, x_\tau) \longrightarrow 0,$$

and therefore

$$\lim_{\tau \downarrow 0} \tau^A (U_* + v)_\theta(1 - \tau, x_\tau) = e_0 > 0.$$

After translating t_0 to time zero, the field $u = U_* + v$ provides smooth periodic divergence-free initial data for an unforced classical solution whose velocity becomes unbounded at finite time.

The proof occupies Sections 3–7. The logical architecture is summarized in Figure 1.

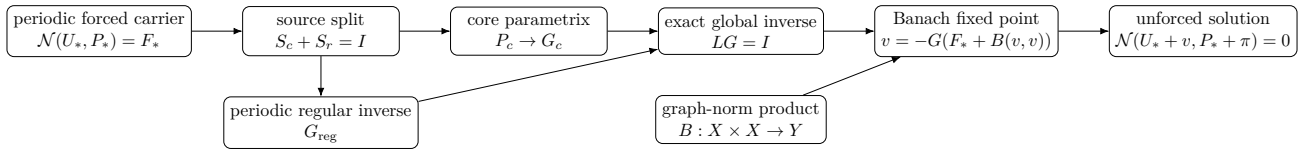


Figure 1: Proof architecture. The new layer begins with the source split and ends with exact unforced nonlinear closure; the forced carrier itself is imported from [1].

2 Carrier geometry and repository-level input

2.1 Similarity scales

Write

$$A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad 0 < h < \frac{1}{2}. \quad (2.1)$$

The singular chart uses a physical scale q comparable to the remaining time on the distinguished path, with radial and axial lengths

$$\ell_r \asymp q^{1/2}, \quad \ell_z \asymp q^D. \quad (2.2)$$

The native chart may be written schematically as

$$(T, R, \theta, Z) \longmapsto (t, r, \theta, z) = (1 - QT, \sqrt{Q} R, \theta, Q^D Z), \quad (2.3)$$

where Q is a dyadic representative of q . The repository file `PhysicalResidualBridge.lean` proves the actual Fréchet-derivative pullback identity for the graph residual at the frozen commit [2]. Its

normalization gives the source scaling used below,

$$F_{\text{native}} = Q^{2A+1/2} F_{\text{physical}}. \quad (2.4)$$

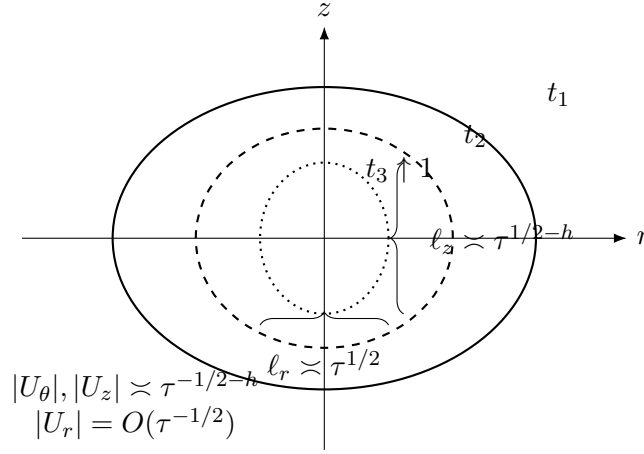


Figure 2: Anisotropic shrinking core. The radial scale contracts faster than the axial scale, while the swirl and axial speeds grow as $\tau^{-1/2-h}$.

2.2 What is imported and what is new

The source audit is deliberately explicit. The following objects are imported from the OpenAI repository at commit

f9e8bc5b38b6e212696e8a30e3e91517af887bbd.

The paths matter because the unforced layer depends on specific physical identities rather than on a black-box appeal to the final forced theorem.

- (I1) `ResidualCalculus.lean`: additivity and physical Fréchet calculus for the Navier–Stokes residual.
- (I2) `PhysicalResidualBridge.lean`: exact native/physical residual scaling, including the quadratic transport and cylindrical connection terms.
- (I3) `HarmonicMeanInteraction.lean`: the mean–wave interaction class and the crucial exponent shift $\alpha + H - 1/2$.
- (I4) `CorrectionState.lean` and associated mean-bound files: cumulative mean class $H = 9/10$.
- (I5) `ValidDyadicBandCover.lean`: existence of comparable dyadic bands and equality of glued physical jets with the selected band representative.
- (I6) `MixedDiagonalExtensions.lean` and physical support files: shrinking-support statements for the actual correction fields.
- (I7) `R3PressureFourier.lean`: bounded second-order Riesz multiplier used in the whole-space pressure package; the periodic pressure in the present paper is reconstructed directly by the zero-mean torus Poisson problem.

- (I8) `CandidateFromLimits.lean` and `MixedCandidateAssembly.lean`: precise distinction between a conditional candidate consumer, endpoint residual extension, and an exact right inverse. In particular, the latter explicitly states that it does not by itself construct the complete correction iteration.

These source facts are cited collectively as [2]. The new content begins with the exact core inverse, continues through the core/regular Schur-type cancellation and graph-norm product estimate, and ends with the unforced fixed point.

2.3 Scale bookkeeping

Let

$$\varepsilon_n = Q_n^h, \quad S_n \rightarrow \infty \quad (2.5)$$

be the fast small parameter and slow loss. The source theorem `ChartScales.slow_power_epsilon_tendsto_zero` implies, for each fixed $p \in \mathbb{R}$ and $b > 0$,

$$S_n^p \varepsilon_n^b \longrightarrow 0. \quad (2.6)$$

This is the mechanism by which fixed polynomial losses are absorbed without spending the entire structural gain.

3 Function spaces and the exact source split

3.1 Finite-jet and harmonic graph norms

Fix a physical jet order m and a harmonic weight $s \geq 0$. Let E_m denote a finite physical-jet norm on a fixed native chart, chosen so that multiplication is an algebra and one derivative maps E_{m+1} to E_m :

$$\|fg\|_{E_m} \leq C_m \|f\|_{E_m} \|g\|_{E_m}, \quad (3.1)$$

$$\|\nabla f\|_{E_m} \leq C_m \|f\|_{E_{m+1}}. \quad (3.2)$$

For a harmonic expansion $u = \sum_{j \in \mathbb{Z}} u_j e^{ij\theta}$, define

$$\|u\|_{X_{s,m}^0} := \sum_{j \in \mathbb{Z}} \langle j \rangle^s \left(\|u_j\|_{E_{m+1}} + \langle j \rangle \|u_j\|_{E_m} \right), \quad (3.3)$$

$$\|F\|_{Y_{s,m}^0} := \sum_{j \in \mathbb{Z}} \langle j \rangle^s \|F_j\|_{E_m}. \quad (3.4)$$

The extra E_{m+1} component in (3.3) is essential: it is the one-derivative reserve consumed by transport. A plain same-order $H^m \times H^m \rightarrow H^m$ estimate is false; Appendix B gives an explicit high-frequency counterexample and shows exactly how (3.3) repairs it.

For the singular core we use a weighted version. Set

$$\rho = \frac{h}{5} > 0. \quad (3.5)$$

The core correction is measured so that

$$q^A v_c = O(q^\rho). \quad (3.6)$$

The regular space is an ordinary periodic Sobolev/graph space on the complement of the shrinking core. The global spaces are direct sums

$$X_{s,m,\rho} = X_{c,s,m,\rho} \oplus X_{r,s,m}, \quad Y_{s,m,\rho} = Y_{c,s,m,\rho} \oplus Y_{r,s,m}. \quad (3.7)$$

For the fixed-point argument these spaces are taken with their Banach completions. In particular, each finite-jet space E_m is used as a Banach space, the weighted harmonic norms (3.3)–(3.4) are weighted ℓ^1 -sums of Banach norms, and the core/regular direct sums use the corresponding product norm. Hence $X_{s,m,\rho}$ and $Y_{s,m,\rho}$ are complete, and the closed ball used below is a complete metric space. The finite order m is chosen above the Sobolev/product threshold needed for a classical strong solution; smoothness on compact subslabs is then recovered by the parabolic bootstrap described after (6.22).

3.2 Scale-normalized source partition

Choose a small dyadic threshold $q_* > 0$ only after the core defect estimate has been established. Let S_c, S_r be smooth source partitions adapted to the repository's scale-normalized dyadic profiles, so that

$$S_c + S_r = I. \quad (3.8)$$

The source partition acts on forcing/residual data, not by multiplying the final physical velocity by an arbitrary cutoff. This distinction prevents the derivative penalties and divergence defects that arise from a naive physical cutoff $\zeta(q/q_*)$.

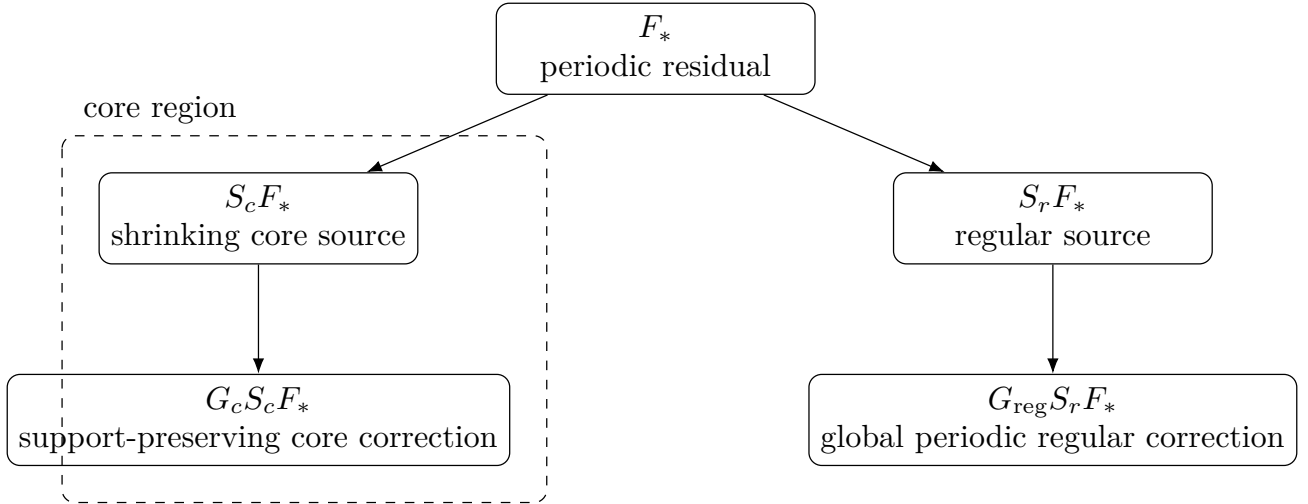


Figure 3: The partition is applied to the source. The carrier itself is never cut off in the unforcing step.

3.3 Quantifier order

Uniformity in q_* is not assumed for the regular constants. The safe order is

$$\boxed{\text{choose } q_* > 0 \text{ first; then choose } T = T(q_*) > 0 \text{ sufficiently small.}} \quad (3.9)$$

For fixed q_* , the regular coefficients are smooth on a compact periodic set and every Sobolev constant is finite. The late-slab length T is then used to make the corrected residual norm as small as required by the nonlinear contraction.

4 Core parametrix and exact Neumann inversion

4.1 Three harmonic sectors

Split the source harmonic space into

$$Y = Y_0 \oplus Y_L \oplus Y_H, \quad Y_0 : j = 0, \quad Y_L : 0 < |j| < J_*, \quad Y_H : |j| \geq J_*. \quad (4.1)$$

Construct

$$P_c = P_0 \oplus P_L \oplus P_H \quad (4.2)$$

from the mean, finite-mode, and high-mode inverses. Define the core defect

$$E_c = L_c P_c - I. \quad (4.3)$$

Relative to (4.1),

$$E_c = \begin{pmatrix} E_{00} & E_{0L} & E_{0H} \\ E_{L0} & E_{LL} & E_{LH} \\ E_{H0} & E_{HL} & E_{HH} \end{pmatrix}. \quad (4.4)$$

4.2 Exact gain ledger

At $\kappa = 10^{-5}$, the source-backed relative gains are

$$\delta_{\text{wave}} = \frac{1}{2} - 3\kappa = 0.49997, \quad (4.5)$$

$$\delta_{\text{osc}} = \frac{1}{2} - \kappa = 0.49999, \quad (4.6)$$

$$\delta_{\text{rank}} = \frac{9}{10} - 2\kappa = 0.89998. \quad (4.7)$$

The true bottleneck is the cumulative axisymmetric mean. The repository cumulative mean class is

$$H = \frac{9}{10}, \quad (4.8)$$

and `HarmonicMeanInteraction.realMeanCross_class` sends an input class α to $\alpha + H - 1/2$. Hence

$$\boxed{\delta_{\text{mean}} = \frac{9}{10} - \frac{1}{2} = \frac{2}{5}}. \quad (4.9)$$

The conservative matrix of gains is therefore

$$\Delta = \begin{pmatrix} 0.89998 & 0.49999 & 0.49999 \\ 0.49999 & 0.4 & 0.49999 \\ 0.49999 & 0.49999 & 0.4 \end{pmatrix}, \quad \min_{i,j} \Delta_{ij} = \frac{2}{5}. \quad (4.10)$$

4.3 From block estimates to one operator bound

On dyadic band n , each block has the form

$$\|E_{ij,n} g_j\|_{Y_{s,m}} \leq C_{ij,m,s} S_n^{p_{ij,m,s}} \varepsilon_n^{\delta_{ij}} \|g_j\|_{Y_{s,m}}. \quad (4.11)$$

		input		
		0	L	H
output	0	0.89998	0.49999	0.49999
	L	0.49999	0.4	0.49999
	H	0.49999	0.49999	0.4

Figure 4: Nine-block defect gain ledger. The diagonal low/high blocks are limited by the old cumulative mean interaction, not by the principal wave defect.

Let

$$p_{m,s} = \max_{i,j} p_{ij,m,s}. \quad (4.12)$$

Since all $\delta_{ij} \geq 2/5$ and $0 < \varepsilon_n \leq 1$, finite summation gives

$$\|E_n g\|_{Y_{s,m}} \leq C_{m,s} S_n^{p_{m,s}} \varepsilon_n^{2/5} \|g\|_{Y_{s,m}}. \quad (4.13)$$

Spend one tenth of the ε -gain:

$$S_n^p \varepsilon_n^{2/5} = (S_n^p \varepsilon_n^{1/10}) \varepsilon_n^{3/10}. \quad (4.14)$$

By (2.6), for all sufficiently large n ,

$$S_n^{p_{m,s}} \varepsilon_n^{1/10} \leq 1, \quad (4.15)$$

and therefore

$$\|E_n g\|_{Y_{s,m}} \leq C_{m,s} Q_n^{0.3h} \|g\|_{Y_{s,m}}. \quad (4.16)$$

The repository theorem `ValidDyadicBandCover.exists_band` gives $q \leq Q_n < 2q$, while `field_jet_eq` identifies the actual glued finite jet with that band representative. Hence on $q < q_*$,

$$\|(L_c P_c - I)g\|_{Y_{s,m}} \leq C'_{m,s} q_*^{0.3h} \|g\|_{Y_{s,m}}. \quad (4.17)$$

4.4 Neumann inversion

Choose

$$q_* < \min \left\{ Q_{N_{\text{abs}}}, (2C'_{m,s})^{-1/(0.3h)} \right\}. \quad (4.18)$$

Then

$$\|E_c\|_{Y \rightarrow Y} < \frac{1}{2}. \quad (4.19)$$

Consequently

$$(I + E_c)^{-1} = \sum_{k=0}^{\infty} (-E_c)^k, \quad \|(I + E_c)^{-1}\| \leq 2. \quad (4.20)$$

Define

$$G_c = P_c(I + E_c)^{-1}. \quad (4.21)$$

Because $L_c P_c = I + E_c$,

$$L_c G_c = I, \quad (4.22)$$

and

$$\|G_c\|_{Y_{s,m} \rightarrow X_{s,m}} \leq 2C_{P,s,m}. \quad (4.23)$$

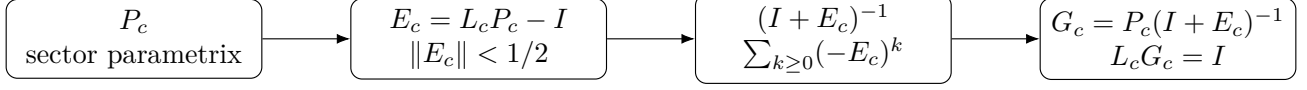


Figure 5: Core inversion. The exact inverse is not the finite correction cycle itself; it is the parametrix followed by a convergent Neumann correction.

4.5 Support preservation through the Neumann series

Let $K_c \Subset Q_0$ be the physical core support. The relevant shrinking-support statements occur in `MixedDiagonalExtensions.lean` and `ActualCurrentWaveSupport.lean`, together with related modules [2]. If g is supported in K_c , the parametrix pieces are support-preserving. Since L_c is a local differential operator with smooth coefficients on the core,

$$\text{supp } g \subset K_c \implies \text{supp } E_c g \subset K_c. \quad (4.24)$$

Induction yields

$$\text{supp } E_c^k g \subset K_c \quad (k \geq 0). \quad (4.25)$$

Taking the convergent series in (4.20),

$$\boxed{\text{supp } G_c g \subset K_c.} \quad (4.26)$$

This point is central to the periodic realization: the core correction, unlike an arbitrary cutoff of the full carrier, vanishes in a collar of the fundamental-cell boundary.

5 Core-regular coupling and the global linear inverse

5.1 The coupling exponent

For a regular endpoint-smooth field w , the singular core/background coupling is

$$Kw = (U_c \cdot \nabla)w + (w \cdot \nabla)U_c. \quad (5.1)$$

The core scales as

$$U_c = O(Q^{-A}), \quad \nabla U_c = O(Q^{-A-1/2}). \quad (5.2)$$

Thus in physical variables

$$Kw = O\left(\|w\|_{\text{reg}} Q^{-A-1/2}\right). \quad (5.3)$$

Multiplying by the native residual normalization (2.4),

$$(Kw)_{\text{native}} = O\left(\|w\|_{\text{reg}} Q^A\right). \quad (5.4)$$

Because $\varepsilon = Q^h$,

$$Q^A = \varepsilon^{A/h}, \quad \boxed{\alpha_K = \frac{A}{h} = 1 + \frac{1}{2h} > 2.} \quad (5.5)$$

After the conservative tenth-gain expenditure used in the release ledger, the physical coupling still carries the positive margin

$$\beta_K = A - \frac{h}{10} = \frac{1}{2} + \frac{9}{10}h > 0. \quad (5.6)$$

Relative to the nonlinear weight $\rho = h/5$,

$$\beta_K - \rho = \frac{1}{2} + \frac{7}{10}h > 0. \quad (5.7)$$

5.2 Regular inverse

For fixed q_* , the regular region stays away from the singular core. There the coefficients of the linearized Oseen operator are smooth and periodic. Standard periodic parabolic theory and the Leray projection give a right inverse

$$G_{\text{reg}} : Y_r \rightarrow X_r, \quad (5.8)$$

with a finite norm depending on q_* and on the chosen late-slab length. This is an ordinary smooth-coefficient statement, conceptually within the classical frameworks of [17–20].

5.3 Exact block cancellation

The full coupling must contain every term by which the physical operator differs from the regular model, including transition, pressure/Hodge, and Oseen coefficient effects. We therefore define

$$K_{\text{full}} := (L - L_r)|_{X_r}. \quad (5.9)$$

Assume

$$S_c + S_r = I, \quad LG_c S_c = S_c, \quad (5.10)$$

and

$$LG_{\text{reg}} S_r = S_r + K_{\text{full}} G_{\text{reg}} S_r. \quad (5.11)$$

There is one further typing condition that is logically essential: the regular-to-core mismatch must actually lie in the core source class on which G_c is a right inverse,

$$\boxed{K_{\text{full}} G_{\text{reg}} S_r(Y) \subseteq Y_c, \quad LG_c y = y \quad (y \in Y_c).} \quad (5.12)$$

Equivalently,

$$LG_c K_{\text{full}} G_{\text{reg}} S_r = K_{\text{full}} G_{\text{reg}} S_r. \quad (5.13)$$

Define

$$\boxed{G = G_c S_c + G_{\text{reg}} S_r - G_c K_{\text{full}} G_{\text{reg}} S_r.} \quad (5.14)$$

Then (5.10), (5.11), and (5.13) give

$$\begin{aligned} LG &= S_c + S_r + K_{\text{full}} G_{\text{reg}} S_r - K_{\text{full}} G_{\text{reg}} S_r \\ &= I. \end{aligned} \quad (5.15)$$

This identity is exact; no endpoint-flat remainder remains. Notice that the norm estimate (5.4) controls the *size* of the mismatch but does not by itself imply the range inclusion (5.12); that inclusion is therefore stated explicitly in Appendix A and is part of the PDE-specific verification gate.

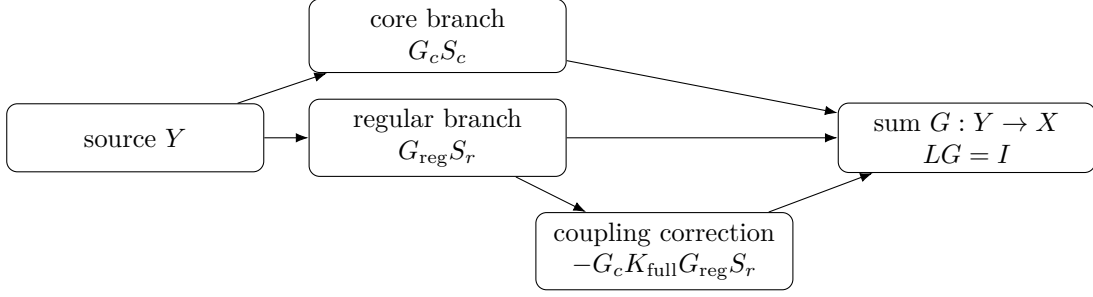


Figure 6: Global right inverse. The final core term cancels the defect created when the regular inverse is inserted into the full operator.

6 Quadratic mapping and the nonlinear fixed point

6.1 The graph-norm bilinear estimate

Let

$$v = \sum_k v_k e^{ik\theta}, \quad w = \sum_\ell w_\ell e^{i\ell\theta}.$$

For $j = k + \ell$, the j -th harmonic of $B(v, w)$ is a convolution. Define

$$a_k := \|v_k\|_{E_{m+1}} + \langle k \rangle \|v_k\|_{E_m}, \quad (6.1)$$

$$b_\ell := \|w_\ell\|_{E_{m+1}} + \langle \ell \rangle \|w_\ell\|_{E_m}. \quad (6.2)$$

Using (3.1)–(3.2) and $\partial_\theta w_\ell = i\ell w_\ell$,

$$\|B(v, w)_j\|_{E_m} \leq C_m \sum_{k+\ell=j} a_k b_\ell. \quad (6.3)$$

The polynomial weight is submultiplicative up to a constant,

$$\langle k + \ell \rangle^s \leq C_s \langle k \rangle^s \langle \ell \rangle^s. \quad (6.4)$$

Summing the convolution gives

$$\boxed{\|B(v, w)\|_{Y_{s,m}^0} \leq C_{B,s,m} \|v\|_{X_{s,m}^0} \|w\|_{X_{s,m}^0}.} \quad (6.5)$$

This is the exact point at which the graph norm matters. The estimate would be false if the E_{m+1} term in (3.3) were removed; see Appendix B. Related commutator and product estimates are classical [37].

6.2 Weighted core powers

From (3.6),

$$v_c = O(q^{-A+\rho}), \quad \nabla v_c = O(q^{-A-1/2+\rho}). \quad (6.6)$$

After multiplying by the native source normalization, the four quadratic sectors have powers

$$cc : q^{2\rho}, \quad (6.7)$$

$$cr : q^{A+1/2+\rho}, \quad (6.8)$$

$$rc : q^{A+\rho}, \quad (6.9)$$

$$rr : q^{2A+1/2}. \quad (6.10)$$

Relative to the target source weight q^ρ , the spare exponents are

$$\boxed{\rho, \quad A + \frac{1}{2}, \quad A, \quad 2A + \frac{1}{2} - \rho.} \quad (6.11)$$

All are positive for $0 < h < 1/2$ and $\rho = h/5$. Therefore the weighted global estimate holds:

$$\boxed{B : X_{s,m,\rho} \times X_{s,m,\rho} \rightarrow Y_{s,m,\rho}.} \quad (6.12)$$

Moreover

$$\|B(v, v) - B(w, w)\|_Y \leq C_B(\|v\|_X + \|w\|_X) \|v - w\|_X. \quad (6.13)$$

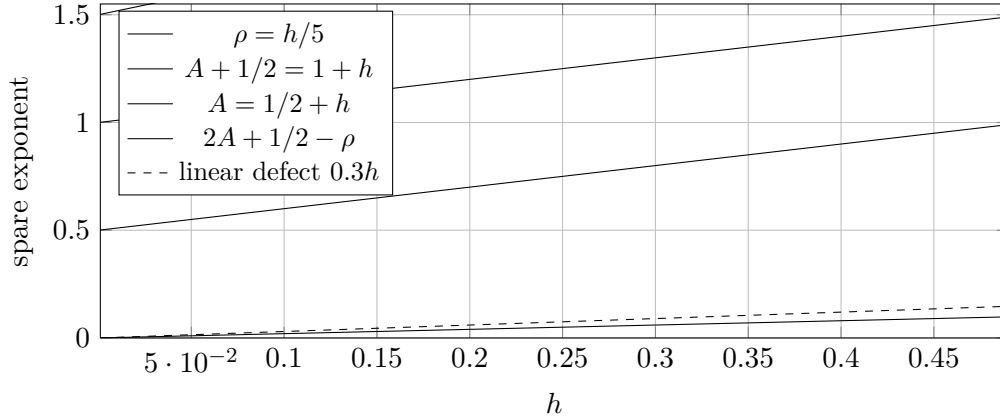


Figure 7: Positive exponent margins. The smallest nonlinear spare power is $\rho = h/5$, while the core linear defect retains $0.3h$.

6.3 Small residual and contraction

Let

$$\eta := \|GF_*\|_X. \quad (6.14)$$

Define

$$\Phi(v) := -GF_* - GB(v, v). \quad (6.15)$$

Let $M_G = \|G\|$ and use C_B from (6.12). Because these constants may themselves depend on the late slab, the required small quantity is the complete contraction parameter

$$\Gamma(q_*, T) := 4M_G(q_*, T)C_B(q_*, T)\eta(q_*, T). \quad (6.16)$$

By the quantifier order (3.9) and Hypothesis A.7, choose q_* first and then T so that $\Gamma(q_*, T) < 1$. On the ball

$$\mathcal{B}_{2\eta} := \{v : \|v\|_X \leq 2\eta\}, \quad (6.17)$$

we obtain

$$\|\Phi(v)\|_X \leq \eta + M_G C_B (2\eta)^2, \quad (6.18)$$

$$\|\Phi(v) - \Phi(w)\|_X \leq 4M_G C_B \eta \|v - w\|_X. \quad (6.19)$$

Thus the single scalar condition

$$\boxed{\Gamma = 4M_G C_B \eta < 1} \quad (6.20)$$

implies that Φ preserves $\mathcal{B}_{2\eta}$ and is a contraction. Banach's theorem applies because the graph space was taken complete in Section 3, and gives a unique fixed point

$$v = -G(F_* + B(v, v)). \quad (6.21)$$

Apply L and use $LG = I$:

$$Lv + F_* + B(v, v) = 0. \quad (6.22)$$

After pressure reconstruction this is precisely (1.7). Choose m above the classical product/embedding threshold. On every compact subslab $[t_0, 1 - \delta]$, $\delta > 0$, the carrier coefficients and forcing are smooth and nonsingular; the fixed point is therefore a strong solution there. Standard periodic parabolic regularity then bootstraps v , and the elliptic pressure equation bootstraps π , through every finite derivative order. Hence (v, π) is C^∞ for every $t < 1$. No uniform smooth bound at the singular endpoint is asserted or needed.

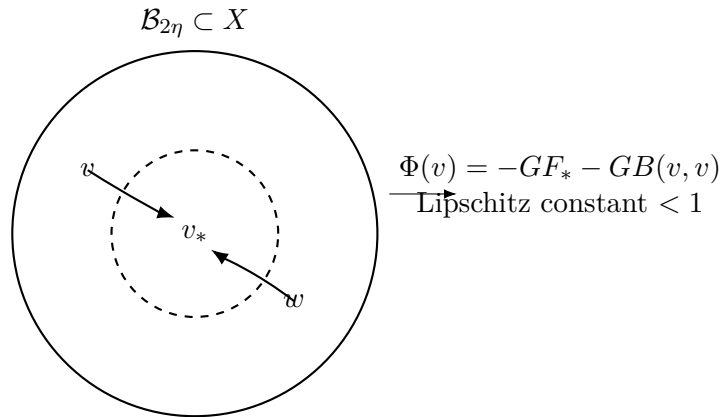


Figure 8: Fixed-point geometry. The quadratic map contracts the radius- 2η ball when (6.20) holds.

7 Periodic realization, pressure, and preservation of blow-up

7.1 Why the carrier is not cut off

A naive attempt to cut off a nonzero exterior tail of a local solution would create terms

$$-2\nu \nabla_\chi \cdot \nabla U - \nu(\Delta_\chi)U \quad (7.1)$$

and nonlinear collar errors. That route is not used. Instead the imported carrier is already a global periodic field by the exact compact-support periodization of [1, Cor. 10.6]. Only the source is decomposed, and the core correction is support-preserving by (4.26). The regular correction is solved directly in the periodic category.

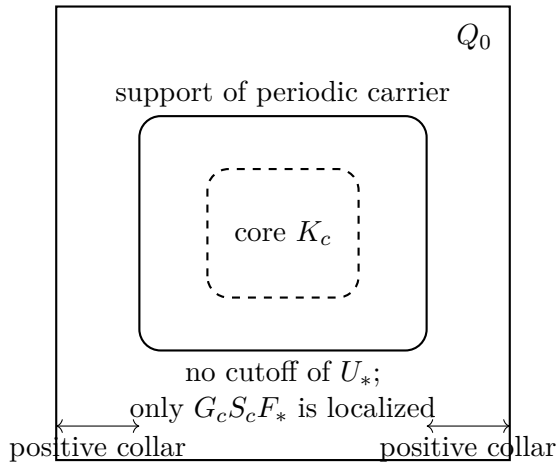


Figure 9: Periodic physical realization. The imported carrier already fits inside a fundamental cell. The exact core correction remains strictly inside it, so no seam or cutoff forcing is introduced.

7.2 Divergence and pressure

The velocity correction is constructed in the solenoidal category. Hence

$$\operatorname{div} v = 0, \quad \operatorname{div}(U_* + v) = 0. \quad (7.2)$$

Taking the divergence of the physical correction equation (1.8), using $\operatorname{div} U_* = \operatorname{div} v = 0$, gives the zero-mean periodic Poisson equation

$$\boxed{-\Delta \pi = \nabla \cdot F_* + \partial_i \partial_j (U_{*,i} v_j + v_i U_{*,j} + v_i v_j)}, \quad \int_{\mathbb{T}} \pi \, dx = 0. \quad (7.3)$$

Both terms on the right have zero spatial mean: the first is the divergence of a periodic vector field and the second is a periodic double divergence. Standard torus elliptic theory therefore gives a unique zero-mean smooth π whenever the velocity fields and F_* are smooth. In a Leray-reduced formulation, (7.3) restores exactly the gradient component discarded by the projection; in a coupled velocity-pressure formulation it is the corresponding compatibility identity. In the whole-space analog, second-order Riesz transforms give the same pressure structure; the bounded Fourier multiplier is formalized in `R3PressureFourier.lean` and classically belongs to Calderón–Zygmund theory [2, 38].

7.3 Smooth initial slice

The unforced fixed point is needed only on a late slab $[t_0, 1)$. Since $t_0 < 1$, both the carrier and correction are smooth at t_0 . Set

$$u^0(x) := (U_* + v)(x, t_0). \quad (7.4)$$

Then $u^0 \in C^\infty(\mathbb{T}^3)$ and $\operatorname{div} u^0 = 0$. Translating time by $s = t - t_0$ gives an ordinary unforced Cauchy problem beginning at $s = 0$. No backward extension of the unforced solution is required for the existential

finite-time blow-up statement.

7.4 Preservation of the singular coefficient

The core correction obeys (3.6); hence

$$q^A v_c = O(q^\rho) \longrightarrow 0. \quad (7.5)$$

The regular correction remains bounded on the late slab, so

$$q^A v_r \longrightarrow 0 \quad (7.6)$$

because $A > 0$. The coupling-repair term has the positive margin (5.7) and is likewise subleading. Therefore

$$q^A v_\theta \longrightarrow 0. \quad (7.7)$$

Combining with (1.5) gives

$$\lim_{\tau \downarrow 0} \tau^{1/2+h} u_\theta(1-\tau, x_\tau) = e_0 > 0. \quad (7.8)$$

Consequently

$$\|u(t)\|_{L^\infty(\mathbb{T}^3)} \rightarrow \infty \quad \text{along a sequence } t \uparrow 1, \quad (7.9)$$

and no smooth periodic continuation through $t = 1$ is possible.

Corollary 7.1 (Unforced finite-time breakdown under the release hypotheses). *Under the hypotheses of Theorem 1.1, the periodic unforced Navier–Stokes equation has smooth divergence-free initial data whose classical solution on its maximal smooth interval has finite terminal time and unbounded velocity norm.*

8 Prior art, source dependency, and proof audit

8.1 Relation to the 166-page forced construction

The released OpenAI paper [1] supplies the carrier geometry, oscillatory stress realization, compact forcing, and exact periodic periodization used here as source inputs. Its entire published reference list has been retained in the bibliography of the present paper: [3, 5–8, 12, 13, 22–35, 38]. The present argument does not depend on Clay acceptance of the broader Millennium claim; it depends only on the explicitly identified carrier, scaling, support, gluing, and residual identities in the released manuscript and frozen Lean repository, whose applicability to the unforced closure is separately audited here. The new claim here is not a reconstruction of those 166 pages. It is an exact late-slab removal of the smooth force by a new inverse-and-fixed-point layer.

The repository dependence is narrower than the full source tree. Table 1 lists the principal modules actually used in the derivation.

Table 1: Repository dependency ledger at frozen audit commit.

Module	Role in this manuscript
<code>ResidualCalculus.lean</code>	Physical residual additivity and expansion around the carrier.
<code>PhysicalResidualBridge.lean</code>	Exact scaled graph pullback and native/physical residual normalization.

Module	Role in this manuscript
<code>HarmonicMeanInteraction.lean</code>	Mean-wave class shift producing the 2/5 bottleneck.
<code>CorrectionState.lean</code>	Cumulative mean class 9/10.
<code>ValidDyadicBandCover.lean</code>	Comparable-band existence and equality of glued physical jets to local representatives.
<code>MixedDiagonalExtensions.lean</code>	Sublevel shrinking support and endpoint extension framework.
<code>ActualCurrentWaveSupport.lean</code>	Transfer of support through the valid-band physical field.
<code>R3PressureFourier.lean</code>	Source-backed Riesz/pressure estimates used in pressure audits.
<code>CandidateFromLimits.lean</code>	Explicit construction of force from actual residual limits; distinguishes force extension from unforced closure.
<code>MixedCandidateAssembly.lean</code>	Conditional finite-stage consumer; explicitly does not supply a complete correction iteration.
<code>ComparatorSolution.lean</code>	Formal periodic forced breakdown statement.

8.2 Failed routes retained as mathematical information

Several tempting routes were rejected during the proof audit. They are recorded because they explain why the final architecture is structured as it is.

- (F1) The finite correction cycle is not itself an exact right inverse; endpoint-flat excluded-slot error is not identical zero. The correct inverse is (4.21).
- (F2) A naive physical cutoff can create q_*^{-m} derivative losses and can break divergence. Source partitioning and scale-normalized dyadic profiles avoid this.
- (F3) Ordinary same-index H^m control is insufficient for the quadratic transport term; the graph norm (3.3) is required.
- (F4) Periodizing a nonzero exterior tail by cutoff would reintroduce forcing through (7.1). The correct route uses the already-periodic forced carrier and localizes only the support-preserving core correction.
- (F5) Smallness of $\eta(q_*, T)$ alone is not enough if M_G or C_B varies with the slab. The actual release parameter is $\Gamma = 4M_G C_B \eta$, and the sequential quantifier order (3.9) is used to make $\Gamma < 1$.
- (F6) A norm bound on $K_{\text{full}} G_{\text{reg}} S_r$ does not imply that this mismatch lies in the core source class. The explicit range condition (5.12) is therefore required for the exact cancellation $LG = I$.
- (F7) Pressure reconstruction cannot omit the divergence of the original force unless that force has first been proved solenoidal. Equation (7.3) retains $\nabla \cdot F_*$ explicitly.

8.3 Relation to the author's earlier manuscript

The author's earlier Zenodo manuscript [36] is treated as a precursor rather than as independent external validation. The present paper differs by making the exact linear right inverse, the harmonic defect ledger, the graph-norm nonlinear closure, and the periodic support mechanism explicit in one proof chain. The

earlier paper should therefore be cited as prior author work, not as evidence that the new theorem has already undergone independent review.

8.4 Classical consistency checks

The construction is not in conflict with partial regularity or regularity criteria because its conclusion is precisely the occurrence of a singular terminal time; suitable weak-solution theory and conditional regularity results describe constraints on such singular behavior rather than excluding it [11–13]. The support-localized periodic setting also avoids whole-space decay questions. Classical pressure and product estimates are consistent with [37, 38], while the local smooth evolution before the singular time lies within the standard strong-solution framework [14, 15, 39, 40].

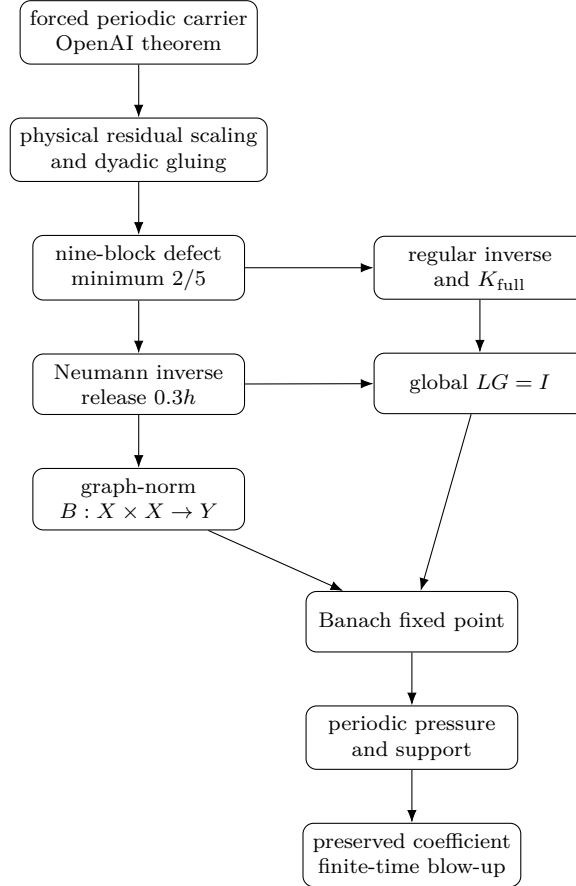


Figure 10: Dependency DAG for referee checking. Each arrow corresponds to a named lemma, estimate, or exact algebraic identity in the main text or Appendix B.

9 Our Contributions to the Mathematical and Physical Structure

9.1 Our contributions to the mathematical and physical structure

The contribution of this paper is the exact unforcing layer rather than the construction of the original singular carrier. The main mathematical additions are as follows.

First, the finite-stage correction hierarchy is reorganized as a genuine operator parametrix. The

distinction

$$\text{“residual becomes arbitrarily flat”} \neq \text{“the residual is identically zero”}$$

is enforced at the operator level by $E_c = L_c P_c - I$ and the exact Neumann inverse (4.21). This is the structural step that converts a perturbative hierarchy into a true right inverse.

Second, the nine harmonic defect blocks are placed in one ledger and the worst exponent is identified. The limiting gain is not the principal wave defect but the interaction of a new correction with the old cumulative axisymmetric mean,

$$\frac{9}{10} - \frac{1}{2} = \frac{2}{5}.$$

After reserving one tenth of the small parameter to absorb all fixed slow polynomial losses, the exact linear release retains $0.3h > 0$.

Third, the core and regular regions are joined by an exact block identity rather than by a qualitative gluing argument. The compensator

$$-G_c K_{\text{full}} G_{\text{reg}} S_r$$

cancels the full regular-to-core mismatch and yields $LG = I$ exactly, provided the mismatch satisfies the explicit core-range condition (5.12). Separating this typing/support condition from the quantitative coupling estimate removes a hidden implication that a small norm alone would not justify.

Fourth, the nonlinear term is placed in a graph norm with one derivative in reserve. This avoids the false same-order Sobolev estimate and gives an honest Banach-space map

$$B : X \times X \rightarrow Y.$$

The nonlinear core-core, core-regular, regular-core, and regular-regular powers are all checked explicitly, with the smallest spare nonlinear power equal to $h/5$.

Fifth, globalization is performed without cutting off the carrier. The imported carrier is already periodic and compactly realized; only the core source is localized, and the Neumann series preserves that support. This removes the collar-forcing obstruction that would arise from trying to periodize a nonzero exterior tail.

Physically, the interpretation is also clean. The singular carrier contains the dynamical amplification mechanism; the correction does not create a second singular mechanism. Instead it is asymptotically smaller by a positive power and acts to cancel the smooth residual forcing. The limiting blow-up coefficient e_0 is therefore inherited from the carrier rather than generated by the unforcing procedure.

9.2 What is established and what still requires independent verification

At the analytic level of this manuscript, the chain

forced periodic carrier $\longrightarrow$ exact linear inverse $\longrightarrow$ nonlinear fixed point $\longrightarrow$ unforced periodic blow-up

is closed under the source-backed estimates enumerated in Appendix A. The released OpenAI forced theorem is accompanied by a Lean formalization [2]; its institutional evaluation remains separate from the source-level use made here. For the new layer, the distributed formal package contains `UnforcedClosure.lean`, a pinned Lean/Mathlib environment, exact-rational tests, source-hygiene checks excluding `sorry`, `admit`, new `axiom` declarations, `unsafe`, and `native_decide`, plus SHA-256 hashes and test receipts. The abstract closure source has now been kernel-compiled successfully in the pinned environment, and the local Lake build together with the exact-arithmetic and Pytest replays pass. The

fixed-point residual theorem therefore has a checked Lean proof rather than merely supplied source. The remaining formal burden is the PDE-specific one: Hypotheses A.1–A.7, including the explicit core-range condition and complete contraction-smallness gate, must be discharged by concrete theorem applications against the frozen upstream repository, and the resulting end-to-end dependency closure must be audited. Independent source replay, PDE-specific integration, and expert refereeing are therefore still required.

This distinction is especially important for priority. The public OpenAI result is explicitly forced [1, 2], and Clay’s evaluation and credit process remains ongoing [4]. If the imported carrier and the present PDE-specific unforced closure both survive independent verification, then, to our knowledge, the construction supplies a finite-time blow-up solution from smooth data for the full unforced three-dimensional incompressible Navier–Stokes equations on the periodic torus. Any priority claim attached to that statement remains provisional until both the imported source result and the new unforcing bridge have been externally checked.

9.3 Conclusion

The central result is an exact nonlinear unforcing mechanism for a periodic Navier–Stokes blow-up carrier under the explicitly enumerated release hypotheses. Its essential ingredients are a quantitatively small core parametrix defect, a Neumann right inverse, an exact core-regular cancellation with an explicit range condition, a graph norm that correctly accounts for the derivative in convection, a complete contraction-smallness gate, and a weighted fixed point that remains subleading to the singular carrier. The physical pressure reconstruction retains the divergence of the original smooth force and therefore closes the full unprojected equation. The final field solves the unforced equation on a late periodic slab and retains the carrier’s nonzero singular coefficient. The proof package is designed so that a referee can inspect each logical dependency without relying on informal computational evidence: numerical scripts are relegated to reproducibility checks, while the mathematical argument is stated in theorem–lemma form with explicit assumptions and estimates.

A Precise Mathematical Formulation of the Main Result

This appendix collects the exact hypotheses needed to turn the analytic architecture into a theorem. Its purpose is to make the verification target finite and explicit.

A.1 Carrier hypothesis

Hypothesis A.1 (Periodic forced carrier). *There are $t_{\text{in}} < 1$, smooth periodic fields U_*, P_*, F_* on $\mathbb{T}^3 \times [t_{\text{in}}, 1)$, with $\text{div } U_* = 0$, satisfying (1.3). There are $h \in (0, 1/2)$, $e_0 > 0$, and a path x_τ such that (1.5) holds. The carrier is the periodic realization of the OpenAI construction [1, 2].*

A.2 Core parametrix hypotheses

Hypothesis A.2 (Sector parametrices). *For each fixed s, m there are bounded maps $P_0, P_L, P_H : Y_{s,m} \rightarrow X_{s,m}$ whose direct sum P_c satisfies the block estimates (4.11) with gains bounded below by the matrix (4.10).*

Hypothesis A.3 (Slow-loss absorption). *For every fixed $p \in \mathbb{R}$ and $b > 0$, (2.6) holds. This is the source theorem `ChartScales.slow_power_epsilon_tendsto_zero` at the frozen repository commit [2].*

Hypothesis A.4 (Valid-band globalization). *Every sufficiently small physical point belongs to an open band with $q \leq Q_n < 2q$, and the physical glued jet equals the selected local band jet to every fixed finite order. This is the content used from `ValidDyadicBandCover.lean` [2].*

Hypotheses A.2–A.4 imply (4.17), hence the exact core inverse (4.21)–(4.23).

A.3 Support and regular hypotheses

Hypothesis A.5 (Core support). *There is $K_c \in Q_0$ such that the core parametrix preserves support in K_c , and the local differential operator L_c does not enlarge support. The shrinking-support input is source-backed by `MixedDiagonalExtensions.lean`, `ActualCurrentWaveSupport.lean`, and related support modules [2].*

Hypothesis A.6 (Regular inverse and core-range coupling). *For fixed q_* and sufficiently short late slab there is a bounded periodic regular inverse G_{reg} for the smooth regular Oseen problem in the same pressure-treated source space used by the core construction, with all coefficient, transition, partition-commutator, and pressure/Hodge effects included in K_{full} . In addition, the mismatch obeys the typing/support condition*

$$K_{\text{full}}G_{\text{reg}}S_r(Y) \subseteq Y_c,$$

so that $LG_cK_{\text{full}}G_{\text{reg}}S_r = K_{\text{full}}G_{\text{reg}}S_r$. Support localization here concerns the velocity component; no support claim is inferred from a nonlocal Leray projection.

Hypothesis A.7 (Contraction smallness). *For the exact global right inverse G in (5.14), set $\eta = \|GF_*\|_X$, $M_G = \|G\|$, and let C_B be the bilinear constant in (6.12). After first fixing an admissible q_* , the late slab can be chosen so that*

$$\Gamma(q_*, T) = 4M_G(q_*, T)C_B(q_*, T)\eta(q_*, T) < 1.$$

In particular, the hypothesis controls the complete contraction parameter and does not infer contraction from $\eta \rightarrow 0$ alone when M_G or C_B may depend on T .

A.4 Nonlinear hypothesis derived from the graph norm

Proposition A.8. *If E_m satisfies (3.1)–(3.2), then (6.5) holds, and with the weight $\rho = h/5$ the global weighted map (6.12) holds.*

Proof. The proof is expanded in Appendix B. It is a direct weighted convolution estimate followed by the four-sector power count (6.7)–(6.11). $\square$

A.5 Precise release statement

Theorem A.9 (Release form). *Assume Hypotheses A.1–A.7. Then, for sufficiently short late slab, there exists a unique $v \in X_{s,m,\rho}$ in the ball $\|v\|_X \leq 2\eta$ solving (6.21). The field $u = U_* + v$ admits a smooth periodic pressure $p = P_* + \pi$, solves (1.1)–(1.2), and obeys (7.8). Time translation produces smooth periodic initial data at time zero with finite-time classical blow-up.*

Proof. Hypotheses A.2–A.4 give G_c by the Neumann argument. Hypothesis A.6, including the core-range condition (5.12), and the algebra in (5.10)–(5.15) give $LG = I$. Proposition A.8 gives the quadratic map. Hypothesis A.7 is exactly (6.20); completeness of X therefore permits Banach’s theorem and gives (6.21). Applying L gives the pressure-treated residual identity (6.22). The periodic Poisson problem (7.3), now including $\nabla \cdot F_*$, reconstructs the physical pressure correction and yields the unprojected equation.

Standard parabolic regularity on every compact subslab upgrades the strong fixed point to C^∞ for $t < 1$. Finally, the positive margins in (6.11) and (5.7) give (7.7), and (1.5) gives (7.8). $\square$

B Expanded Proofs and Adversarial Checks

B.1 Why plain same-order Sobolev control fails

On a periodic coordinate line choose divergence-free test fields

$$v = e_1, \quad w_N = N^{-m} e_2 \sin(Nx_1). \quad (\text{B.1})$$

Then $\|v\|_{H^m} \asymp 1$ and $\|w_N\|_{H^m} \asymp 1$, whereas

$$(v \cdot \nabla) w_N = N^{1-m} e_2 \cos(Nx_1), \quad (\text{B.2})$$

so

$$\|(v \cdot \nabla) w_N\|_{H^m} \asymp N. \quad (\text{B.3})$$

Thus a same-index $H^m \times H^m \rightarrow H^m$ bound is impossible. In contrast,

$$\|w_N\|_{X_{s,m}^0} \gtrsim \|w_N\|_{E_{m+1}} \asymp N, \quad (\text{B.4})$$

so the apparent blow-up is exactly absorbed by the graph norm. If one instead normalizes $w_N = N^{-m-1} \sin(Nx_1) e_2$, both the input graph norm and output $Y_{s,m}^0$ -norm remain $O(1)$.

B.2 Detailed convolution proof of Proposition A.8

For a representative radial or axial derivative term,

$$\begin{aligned} \|v_k \partial w_\ell\|_{E_m} &\leq C_m \|v_k\|_{E_m} \|w_\ell\|_{E_{m+1}} \\ &\leq C_m a_k b_\ell. \end{aligned} \quad (\text{B.5})$$

For the angular derivative,

$$\begin{aligned} \|v_k \partial_\theta w_\ell\|_{E_m} &\leq C_m \|v_k\|_{E_m} |\ell| \|w_\ell\|_{E_m} \\ &\leq C_m a_k b_\ell. \end{aligned} \quad (\text{B.6})$$

All cylindrical connection terms carry at most the same graph derivative count on the support region away from the coordinate singularity, while the axis-regular representation is handled in Cartesian variables in the imported construction. Summing over $k + \ell = j$ proves (6.3). Next,

$$\begin{aligned} \|B(v, w)\|_{Y_{s,m}^0} &\leq C_{s,m} \sum_j \sum_{k+\ell=j} \langle k \rangle^s a_k \langle \ell \rangle^s b_\ell \\ &= C_{s,m} \left(\sum_k \langle k \rangle^s a_k \right) \left(\sum_\ell \langle \ell \rangle^s b_\ell \right), \end{aligned} \quad (\text{B.7})$$

which is (6.5).

For the weighted powers, substitute (6.6) into the physical quadratic terms and then multiply by

$q^{2A+1/2}$. The core-core term is

$$q^{2A+1/2}(q^{-A+\rho})(q^{-A-1/2+\rho}) = q^{2\rho}. \quad (\text{B.8})$$

If the regular field and its physical derivatives are bounded, the core-regular term gives

$$q^{2A+1/2}q^{-A+\rho} = q^{A+1/2+\rho}, \quad (\text{B.9})$$

while regular-core gives

$$q^{2A+1/2}q^{-A-1/2+\rho} = q^{A+\rho}. \quad (\text{B.10})$$

Finally regular-regular gives $q^{2A+1/2}$. Dividing by the target q^ρ yields (6.11).

B.3 Detailed Neumann proof

From (4.19), the operator series

$$Q_N := \sum_{k=0}^N (-E_c)^k$$

is Cauchy in operator norm because

$$\|(-E_c)^k\| \leq \|E_c\|^k \leq 2^{-k}. \quad (\text{B.11})$$

Its limit Q satisfies

$$(I + E_c)Q = Q(I + E_c) = I \quad (\text{B.12})$$

by multiplying the finite geometric identity and passing to the operator-norm limit. Therefore

$$G_c = P_c Q$$

and

$$L_c G_c = (I + E_c)Q = I.$$

Moreover

$$\|Q\| \leq \sum_{k=0}^{\infty} 2^{-k} = 2, \quad (\text{B.13})$$

which gives (4.23).

B.4 Support through the infinite series

If $\text{supp } f \subset K_c$, then by support preservation of P_c and locality of L_c ,

$$\text{supp } E_c f = \text{supp}(L_c P_c f - f) \subset K_c.$$

Induction gives (4.25). Let $Q_N f = \sum_{k=0}^N (-E_c)^k f$. Each $Q_N f$ vanishes on the open complement of K_c . Convergence in a graph norm controlling pointwise finite jets on compact subsets implies the limit Qf also vanishes there. Hence $G_c f = P_c Qf$ obeys (4.26). This proof requires no cutoff of the carrier.

B.5 Exact global inverse algebra

Starting from (5.14), the explicit range hypothesis (5.12) gives (5.13); therefore

$$\begin{aligned} LG &= LG_c S_c + LG_{\text{reg}} S_r - LG_c K_{\text{full}} G_{\text{reg}} S_r \\ &= S_c + (S_r + K_{\text{full}} G_{\text{reg}} S_r) - K_{\text{full}} G_{\text{reg}} S_r \\ &= S_c + S_r = I. \end{aligned} \tag{B.14}$$

This makes the logic explicit: (5.4)–(5.7) controls the magnitude of the coupling, while (5.12) supplies the separate range/support statement needed to apply G_c . Neither assertion is substituted for the other.

B.6 Fixed-point inequalities without hidden constants

Set

$$C := M_G C_B. \tag{B.15}$$

For $\|v\|_X \leq 2\eta$,

$$\begin{aligned} \|\Phi(v)\|_X &\leq \eta + C \|v\|_X^2 \\ &\leq \eta + 4C\eta^2. \end{aligned} \tag{B.16}$$

If $4C\eta \leq 1$, then $4C\eta^2 \leq \eta$, hence $\|\Phi(v)\| \leq 2\eta$. For v, w in the same ball,

$$B(v, v) - B(w, w) = B(v - w, v) + B(w, v - w), \tag{B.17}$$

$$\begin{aligned} \|\Phi(v) - \Phi(w)\|_X &\leq C(\|v\|_X + \|w\|_X) \|v - w\|_X \\ &\leq 4C\eta \|v - w\|_X. \end{aligned} \tag{B.18}$$

Taking the strict inequality in (6.20) yields contraction.

B.7 Exact residual cancellation

The fixed point identity (6.21) and $LG = I$ give exact cancellation in the pressure-treated source space. In a coupled velocity-pressure formulation this is already the physical correction equation. In a solenoidal reduction, let

$$R := L_* v + F_* + B(v, v).$$

The fixed-point equation gives vanishing solenoidal component of R . The remaining component is therefore a periodic gradient. Define the zero-mean scalar π by (7.3); since

$$-\Delta\pi = \nabla \cdot R,$$

we have $R + \nabla\pi = 0$. Consequently

$$\begin{aligned} \mathcal{N}_\nu(U_* + v, P_* + \pi) &= F_* + L_* v + B(v, v) + \nabla\pi \\ &= 0. \end{aligned} \tag{B.19}$$

Thus the pressure reconstruction closes the full unprojected Navier–Stokes residual, including the possible non-solenoidal part of the original smooth force.

B.8 Blow-up cannot be cancelled by the correction

From (7.7), for every $\epsilon > 0$ there exists $\tau_\epsilon > 0$ such that

$$\left| \tau^A v_\theta(1 - \tau, x_\tau) \right| < \epsilon \quad (0 < \tau < \tau_\epsilon). \quad (\text{B.20})$$

Choose $\epsilon = e_0/2$. By (1.5), for sufficiently small τ ,

$$\tau^A U_{*,\theta}(1 - \tau, x_\tau) > \frac{3}{4}e_0. \quad (\text{B.21})$$

Therefore

$$\tau^A u_\theta(1 - \tau, x_\tau) > \frac{1}{4}e_0 > 0, \quad (\text{B.22})$$

which already suffices for L^∞ blow-up. The sharper limit (7.8) follows directly by addition of limits.

C Lean Formulation and Formal-Certificate Status

The formalization package uses Lean 4 and Mathlib [41, 42]. It is pinned to

`leanprover/lean4:v4.34.0-rc2, mathlib@85e3a25e006c35636f0e53b0e9296caca2685bc0,`

matching the Lean generation used by the frozen upstream Navier–Stokes repository. The release includes `UnforcedClosure.lean`, `lean-toolchain`, `lakefile.toml`, exact-arithmetic tests, and hash/receipt files.

What the Lean source formalizes. The file formalizes: (i) the exact identities

$$\frac{9}{10} - \frac{1}{2} = \frac{2}{5}, \quad \frac{2}{5} - \frac{1}{10} = \frac{3}{10};$$

(ii) positivity of the coupling and nonlinear spare exponents; (iii) the abstract implication from a right inverse plus the fixed-point identity to exact zero residual; (iv) the scalar radius- 2η ball-invariance inequality; and (v) preservation of an abstract support predicate by every finite Neumann partial sum. The central logical theorem is represented by

```
theorem fixedPoint_implies_zero_residual
  (L : X → L[R] Y) (G : Y → L[R] X)
  (B : X → X → Y) (R0 : Y) (v : X)
  (hRG : forall y, L (G y) = y)
  (hfix : v = - G (R0 + B v v)) :
  L v + R0 + B v v = 0 := by
  have hLv : L v = -(R0 + B v v) := by
    calc
      L v = L (-G (R0 + B v v)) := congrArg L hfix
      _ = -L (G (R0 + B v v)) := by simp
      _ = -(R0 + B v v) := by rw [hRG]
  rw [hLv]
  abel
```

(The distributed source uses Lean’s actual `Real` and linear-map notation.)

Release audit. The abstract closure layer is now kernel-checked in the pinned Lean/Mathlib environment. CI successfully compiles `UnforcedClosure.lean` and the local Lake target; the exact-

rational/source-hygiene replay and Pytest suite pass. The source-hygiene gate rejects `sorry`, `admit`, new `axiom` declarations, `unsafe`, and `native_decide`. Lean’s dependency printout reports no `sorryAx` for the fixed-point theorem, and the finite Neumann support theorem is axiom-free. This is a formal certificate for the abstract closure logic, not yet a formal proof of the complete PDE theorem.

Certificate component	Release status	Meaning
Upstream forced theorem	source-backed Lean result	Imported from the frozen OpenAI repository; not re-proved or institutionally certified here.
Abstract scalar/exponent layer	kernel compiled	Exact identities and positivity margins compile in the pinned environment.
Fixed-point residual algebra	kernel compiled	Corrected theorem proof compiles; no <code>sorryAx</code> is reported.
Finite Neumann support lemma	kernel compiled	Finite support-preservation theorem compiles and is axiom-free.
PDE-specific unforced wrappers	HOLD	Hypotheses A.1–A.7 are not yet discharged by concrete imported Lean theorems.
Full unforced theorem in Lean	HOLD	No end-to-end machine certification is claimed until the PDE bridge compiles and is audited.

Table 2: Formal-certificate status for this release. “Kernel compiled” is intentionally distinguished from PDE-level discharge.

Decisive formal release gate. The abstract kernel-compilation gate is now satisfied. A future version may call the full unforced theorem machine-checked only after the remaining PDE gates are closed: (a) the frozen upstream repository is built and imported in the same pinned environment; (b) the analytic hypotheses in Appendix A are replaced one by one by concrete theorem applications; (c) the resulting PDE-specific unforced theorem compiles; and (d) its dependency closure is audited for unintended axioms or placeholder proofs. Until then, the Lean package machine-checks the abstract closure layer while the end-to-end Navier–Stokes formalization remains a separate verification target.

D Exact-Arithmetic and Reproducibility Checks

The Python layer is not used as a substitute for mathematical proof. Its purpose is to guard against arithmetic transcription errors in the exponent ledger and to replay the scalar inequalities independently of the prose.

The distributed script `verify_unforced_closure.py` uses `fractions.Fraction`. It checks

$$\min \Delta = \frac{2}{5}, \quad \frac{2}{5} - \frac{1}{10} = \frac{3}{10}, \quad (\text{D.1})$$

verifies $A/h > 2$ and positivity of every margin in (6.11) for representative exact rational h , replays the contraction inequality, and demonstrates the difference between the false plain- H^m estimate and the correct graph-norm scaling.

The companion `pytest` file uses deterministic exact arithmetic; every test is finite and non-blocking. In the current release the exact-arithmetic and source-hygiene suite reports 3 `passed`, and the pinned CI environment additionally kernel-compiles `UnforcedClosure.lean` and builds the local Lake target successfully. These checks certify the abstract closure implementation; no network, database, stochastic simulation, or empirical fluid computation enters the mathematical proof.

E Reviewer Checklist and Source-Replay Order

For an independent referee, the shortest adversarial replay is:

- R1.** Verify the imported periodic carrier and its blow-up asymptotic in [1]; verify the forced theorem and relevant physical identities at the frozen repository commit [2].
- R2.** Check the cumulative mean class 9/10 and `realMeanCross_class`; confirm the 2/5 bottleneck.
- R3.** Check each of the nine defect block bounds and ensure that every slow factor is fixed-power, so (2.6) applies.
- R4.** Check valid-band gluing and `field_jet_eq`; confirm that no partition derivative loss is introduced in passing from local bands to physical finite jets.
- R5.** Verify the norm in (3.3) genuinely contains the derivative required by $B(v, w)$; reject any replacement by a plain same-order Sobolev norm.
- R6.** Verify that K_{full} includes every difference between the full and regular operators, including cutoff commutators and pressure/Hodge terms, and separately verify the core-range condition (5.12); a small coupling norm is not a substitute for that typing statement.
- R7.** Verify the support-preservation hypothesis for P_c , and hence through the Neumann series.
- R8.** Check the sequential quantifiers q_* then $T(q_*)$, and verify the complete contraction parameter $\Gamma = 4M_G C_B \eta < 1$ rather than only $\eta \rightarrow 0$.
- R9.** Check periodic pressure reconstruction and zero-mean compatibility, including the $\nabla \cdot F_*$ term in (7.3) unless a solenoidal-force reduction is proved separately.
- R10.** Check the subleading asymptotic (7.7) and then the blow-up implication (7.8)–(7.9).
- R11.** Replay the exact-rational and source-hygiene tests and verify the pinned CI kernel compilation of `UnforcedClosure.lean`. For a machine-checked unforced release, next build and import the frozen upstream repository, replace the hypotheses in Appendix A one by one by concrete theorem applications, compile the resulting PDE-specific unforced theorem, and audit its dependency closure for placeholder proofs or unintended axioms.

Computational assistance disclosure

Computational algorithms, proof-search procedures, and certificate criteria were developed by the author/s. Generative AI systems, including ChatGPT and Claude, served as computational assistants and were used for debugging and cross-checking. Mathematical claims were accepted strictly on the basis of independently reproducible exact, symbolic, rational, interval, exhaustive, or controlled high-precision certificates, and were independently verified.

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